1. Introduction and Context

Over the last decade, access to housing has become one of the most pressing social problems in Portugal, with house prices growing much faster than wages. According to Eurostat, house prices in Portugal rose by 17.7% year on year in the third quarter of 2025, the second largest increase in the European Union (Eurostat, 2026c). The situation is particularly evident in Lisbon, where a recent comparison of 127 European cities estimated the average rent of a one-bedroom flat in the city centre at €1,331 per month, against an average net monthly wage of €1,343 (Heskin, 2026).

In October 2026, the issue reached a breaking point, and protests that began in Porto spread to Lisbon and other cities, with tent camps in central squares and marches of thousands of people (Goncalves & Pereira, 2026b; Pereira & Agência Lusa, 2026). Similar movements took place in Spain, France, Belgium and Italy (Goncalves & Pereira, 2026a), a sign that the pressure on household budgets goes beyond Portugal and reaches some of the countries where Portuguese emigrants usually settle. Among the main complaints was that wages had not kept pace with housing and living costs, at a time when rents for new leases had nearly doubled since 2017 (Goncalves & Pereira, 2026b). It is precisely this relationship between earnings and rent that the present work sets out to measure.

Faced with these conditions, many people, especially younger workers, see emigration as one of the few ways out. Around 65,000 Portuguese left the country in 2024, a figure close to the levels of the first decade of the century (Vidigal, 2025). Switzerland and Spain were the main destinations, followed, according to the latest available data, by France, Germany, Belgium, the Netherlands and Luxembourg, while the United Kingdom, which led this list before Brexit, received fewer than 3,000 (Vidigal, 2025).

When Portugal is compared with these countries, the comparison is usually made with gross wages, and on that basis emigrating looks like an obvious gain. Yet gross wages leave out much of what shapes daily life, since taxes, rents and everyday expenses vary considerably from one country to another. For someone thinking about leaving, what matters is how much of their working time would be spent on rent and how much money would be left at the end of the month.

As a reference for affordability, this work uses the housing cost overburden rate defined by Eurostat, under which a household is overburdened when housing costs exceed 40% of its disposable income (Eurostat, 2026d). Building on this, the visualization translates rent into workdays and follows the monthly wage from its gross value to what remains after taxes, rent and everyday expenses, in Lisbon and in ten cities of the main destination countries of Portuguese emigration.

2. Research Questions

The context described above leads to three research questions, which guided the design of the visualization. RQ1 is the central question of this work, while RQ2 and RQ3 extend it to the whole monthly budget and to the choice of destination.

- **RQ1**: How many workdays per month does a full-time worker need to pay their share of the rent of a two-bedroom flat in Lisbon, compared with cities in the main destination countries of Portuguese emigration?
- **RQ2**: How much of the monthly wage is left after taxes, rent and everyday expenses in each of these cities?
- **RQ3**: Is the choice of destination by Portuguese emigrants consistent with the countries whose cities leave the most money at the end of the month?

The main question (RQ1) expresses rent in working time rather than in euros, which makes cities with different wage levels directly comparable. The first sub-question (RQ2) widens the view to the whole monthly budget, since a city where rent takes a small share of the wage may still leave little money once taxes and other expenses are paid. The second sub-question (RQ3) brings migration flows into the analysis. If emigration is driven mainly by economic reasons, the countries that receive more Portuguese should be those offering the best financial outcome, while any mismatch suggests that other factors, such as proximity, language and family networks, also play a role. The map with waffle charts addresses RQ1, and the Sankey diagram addresses RQ2 and RQ3.

3. Data and Pre-processing

The visualization combines six datasets from three main sources, OECD, Eurostat, and Observatório da Emigração, covering 11 cities across nine European countries, with Lisbon as the Portuguese reference (Table 1).

Table 1: Datasets used in the project

Dataset	Source	Unit	Level	Year
Average annual gross wages (OECD, 2025a)	OECD	National currency	Country	2025
Income tax and employee social security contributions (OECD, 2025b) - single, no children, average wage	OECD	% of gross wage	Country	2025
Annual exchange rates (Eurostat, 2026b)	Eurostat	National currency per euro	Country	2025
Average monthly rent of a two-bedroom flat (Eurostat, 2026a)	Eurostat	Euro	City	2025

Dataset	Source	Unit	Level	Year
Nominal expenditure per inhabitant (Eurostat, 2025) - food and non-alcoholic beverages, transport, communication	Eurostat	Euro per year	Country	2024
Arrivals of Portuguese nationals (Observatório da Emigração, 2026)	Observatório da Emigração	People	Country	2025 (2024 for France and Spain)

Data preprocessing was carried out in R (R Core Team, 2026) with the readxl (Wickham & Bryan, 2026), dplyr (Wickham, François, et al., 2026) and tidyr (Wickham, Vaughan, et al., 2026) packages. Irrelevant rows and metadata were removed, values were converted to numeric format, country and city names were standardised across sources, and the datasets were merged by country and, in the case of rents, by city. Wages in other currencies were converted into euros with the annual exchange rates, and net wages were obtained by deducting income tax and employee social security contributions from gross earnings.

To ensure comparability, a standard full-time schedule of 40 hours per week and 52 weeks per year (2,080 hours) was assumed, corresponding to eight-hour workdays. Additionally, each worker was assumed to share a two-bedroom apartment with one other person, paying half of the monthly rent, providing a consistent scenario for comparing cities. On this basis, the indicators in Table 2 were computed for each city.

Table 2: Derived indicators

Indicator	Calculation
Net monthly wage	Net annual wage / 12
Net hourly wage	Net annual wage / 2,080
Rent share	Monthly rent / 2
Workdays needed to pay rent	Rent share / net hourly wage / 8
Share of net wage spent on rent	Rent share / net monthly wage × 100
Left after rent	Net monthly wage − rent share
Left at the end of the month	Left after rent − monthly living expenses

Monthly living expenses were obtained by adding up the three expenditure categories (food, transport, communication) and dividing the total by 12. The share of the wage spent on rent was compared with the 40% threshold. Missing values were not imputed. As the United Kingdom has no data in the expenditure series, London is not included in the final after-expenses comparison. Since emigration data exist only at country level, each city was associated with the arrivals recorded for its country, so Bern and Geneva share the same value, as do Berlin and Munich.

Finally, the processed data were checked for missing essential values and exported with writexl (Ooms & Denney, 2026) to a single Excel workbook ("T03_project_data.xlsx") with city-level indicators, country-level information and expenditure by category.

4. Design Choices

The main goal of the design was to make the financial consequences of emigrating from Portugal intuitive and easy to compare, rather than simply presenting wages and living costs. The final visualization (Appendix 1) combines two charts in a single image, built with ggplot2 (Wickham, 2016) and combined with patchwork (Pedersen, 2025). The overall title states the main finding directly and the subtitle sets the scenario, a full-time worker on the national average wage in 2025. The section titles are written as questions that follow the research questions, RQ1 for the map and RQ2 and RQ3 for the Sankey diagram, so the figure reads from top to bottom as a sequence.

The first chart, a map with waffle charts, turns housing costs into something more tangible, the number of days a person needs to work to pay rent. Each city is represented by a card containing a waffle chart, in which every square is one eight-hour workday and every row of five squares is one working week, covering a full working month of about 21.7 days. Counting days of work makes the financial burden easier to understand than a monetary value alone, and because it takes local wages into account, it allows a direct comparison between cities with very different wage levels. These cards are placed over a map of Europe, drawn with sf (Pebesma, 2018) from Natural Earth country boundaries (Massicotte & South, 2026; Natural Earth, n.d.), and linked to their cities by thin lines, which lets the reader associate each value with a specific destination. The map uses the Lambert Azimuthal Equal-Area projection (EPSG:3035), the standard for European statistics, and countries are filled with a sequential beige-to-brown scale of the net hourly wage, which gives context about national wage levels before the reader looks at any card. All legends were gathered in a single horizontal strip below the map.

While the map isolates rent, the Sankey diagram shows how income decreases step by step. Each city is a band that runs through four stages, from the gross wage to the amount left after taxes, rent and everyday expenses. Band width is proportional to euros per person per month, with the same scale in every column, so the narrowing of a band shows directly how much is lost at each step. Cities are sorted by value within each column, which makes bands cross when a city changes position. To the left of these bands, a bar chart shows the number of Portuguese arrivals by country, aligned with the cities of each country and grouped with brackets where a country has two cities. This places migration flows on the same row as the financial outcome of each destination, which is the comparison addressed by RQ3. In the last column, city names and final values are aligned in two columns, so the ranking at the end of the month can be read at a glance.

Colour links the two charts. The 40% affordability threshold serves as a visual reference that lets the reader see instantly which destinations are more financially demanding. Cities where rent stays within 40% of the net wage are shown in green, those where it goes beyond in red, and a light neutral tone shows the workdays left after rent. In the waffle chart, a dashed stepped line marks exactly where the threshold falls in the month, and the Sankey bands keep the same green and red, so the reader learns the code once and can follow a city across the whole figure. The wage map uses a brown scale and the arrival bars are grey, so that both read as background information and neither is confused with the affordability colours. The green and red also differ in lightness, which keeps them distinguishable for readers with the most common forms of colour vision deficiency. The remaining elements were kept simple, so as not to distract from the main comparisons. Lisbon is highlighted in red as the reference point, and the figure uses a single sans-serif typeface, with hierarchy created through size and weight.

More conventional charts were also considered. A bar chart ranking cities by rent share would be easy to read, but it would lose the geographical context and express rent as a percentage. A choropleth map, where countries are coloured according to their rent burden, was set aside because rents are measured by city, and a single colour per country would hide differences such as those between Bern and Geneva. Stacked bars per city could show how the wage is divided, but not how cities change position from one stage to the next, which the crossing bands of the Sankey diagram make visible.

The creativity of the approach lies in combining complementary representations into one story, namely how much people earn, how much they must work to afford housing and what is left. Translating rent into workdays, placing the waffles on a map and linking each destination to the number of Portuguese who move there turns a numerical comparison into a reading of the trade-offs of leaving Portugal.

5. Evolution of the Visualization

The initial idea was to compare housing affordability in Portugal with the main destinations of Portuguese emigration, expressing rent in working time rather than in euros and placing the results on a map of Europe with waffle charts. The first version (Figure 1) focused on young workers under 30 and covered nine cities in 2022, using a 30% income threshold. Because it relied on the hourly wages of workers under 30, which are considerably lower than the national average, the workdays needed to pay rent were much higher than in the final version (Appendix 1), with Lisbon exceeding a full working month.

Below the map, a slope chart showed how the hours needed to pay rent had changed between 2018 and 2022, and a bar chart showed the main destinations of Portuguese emigrants. However, these three charts addressed relatively independent questions and

did not show whether emigrating could actually provide better financial conditions. Wage data for young workers were also only available up to 2022, which made it impossible to describe the current situation. The focus on young people was therefore dropped in favour of the average worker, for whom 2025 data exist, and the threshold was raised to 40% to follow the Eurostat definition of housing cost overburden.

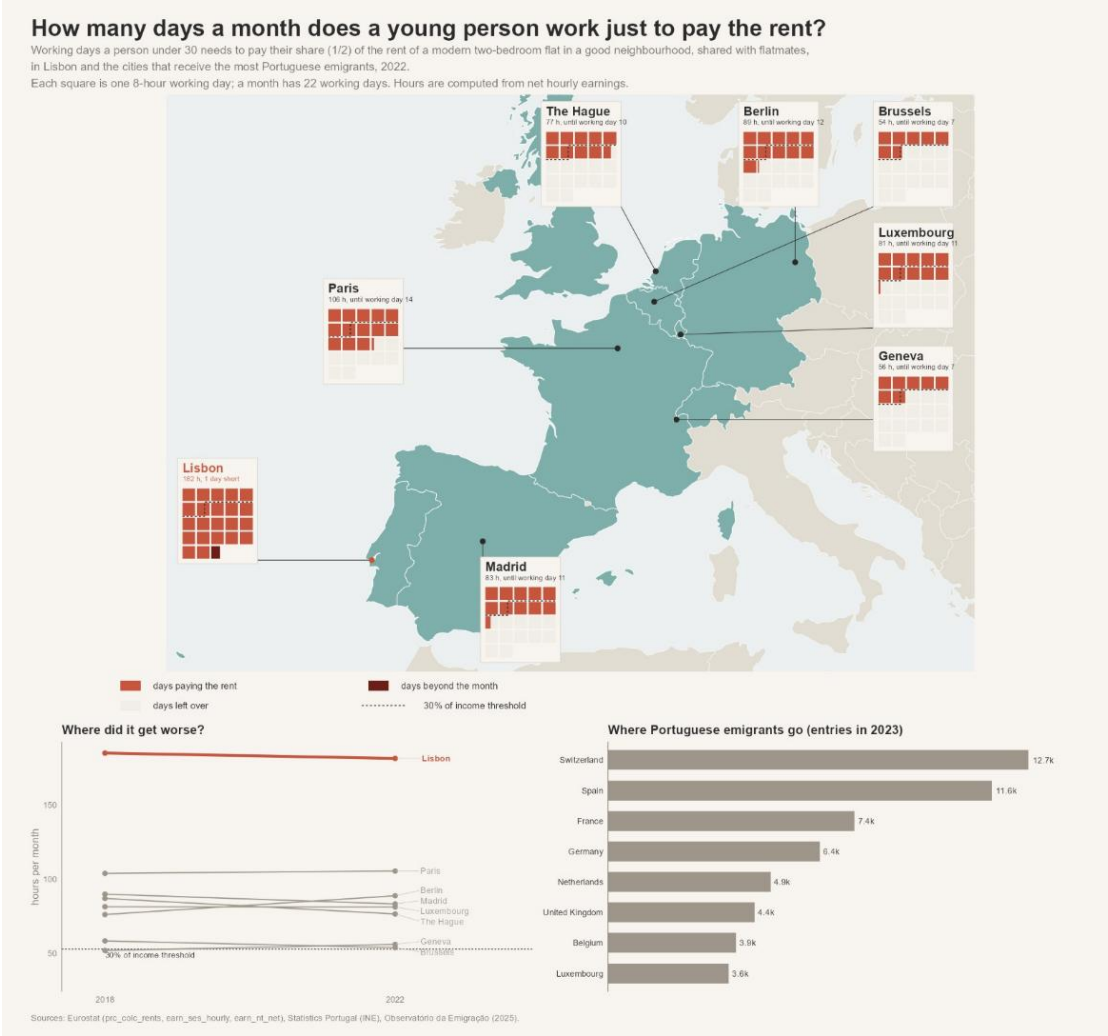


Figure 1: First intermediate version

The main challenge was finding a second visualization that would complement the map without repeating it. The aim was to go beyond housing affordability and look at the financial conditions that shape daily life in each destination. With this in mind, in the second version (Figure 2), the map was extended to eleven cities and its countries were coloured by net hourly wage. The slope chart and the emigration bar chart were replaced by a stacked bar chart showing how gross pay was distributed between taxes, rent and what was left, and a dot plot comparing rent and wage levels in each city with Lisbon as the reference, which introduced taxation into the analysis. Nevertheless, the two lower charts still presented separate indicators, leaving readers to connect the information themselves and in the stacked bars the middle segments were hard to compare between cities. More

importantly, they did not account for everyday living expenses, so the remaining income did not reflect the financial margin a person would actually have in each destination. Emigration had also disappeared from the figure.

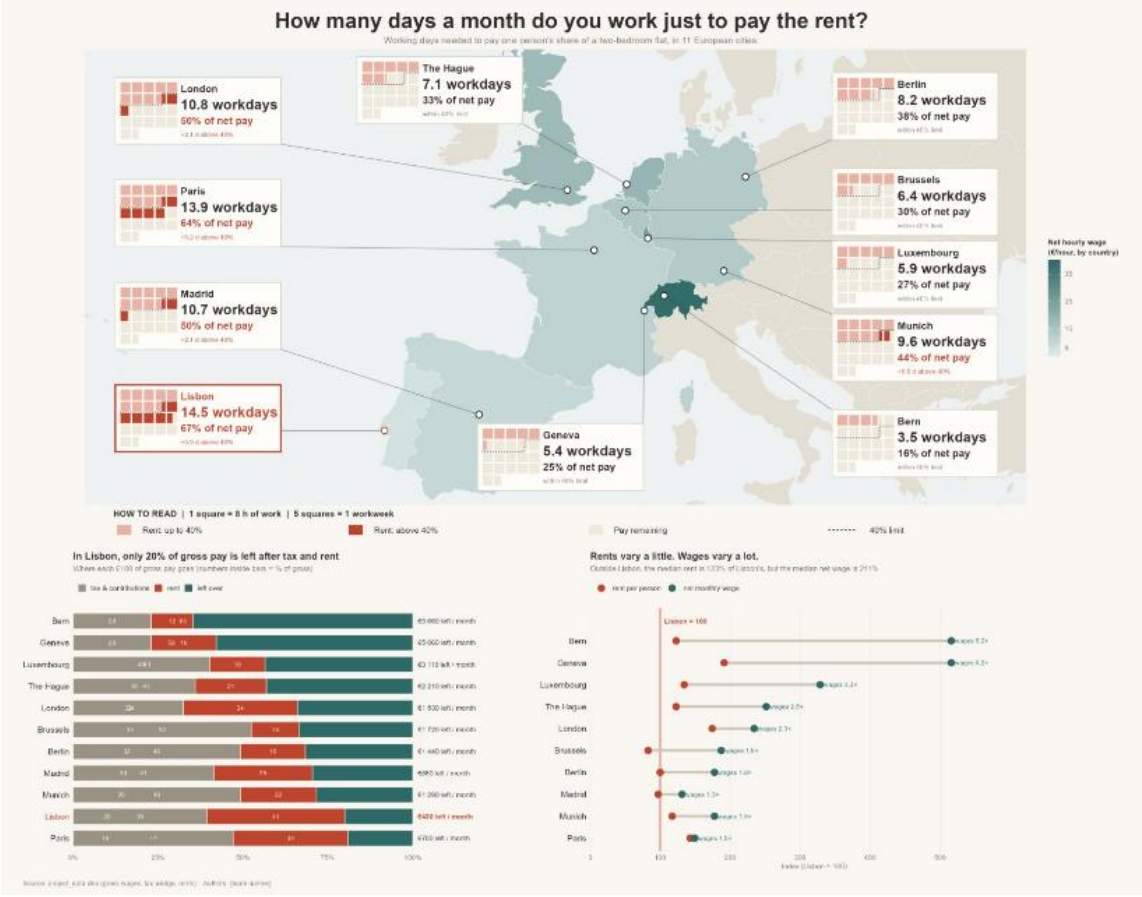


Figure 2: Second intermediate version

At the same time, a revision of the data showed that the OECD indicator used for taxes, the average tax wedge, included employer contributions. It was replaced by the rate of income tax and employee social security contributions (OECD, 2025b), which raised net wages in several countries. In Lisbon, for example, the workdays needed to pay rent fell from 14.5 to 11.7. In the third version (Appendix 2), the map with waffle charts was kept, as the conversion of rent into working days was considered one of the most intuitive and distinctive elements of the project, and its legends and colours were refined. The main change was replacing the two lower charts with a Sankey diagram, which provided the complementary view that had been missing. Portuguese emigration figures were also brought back alongside the Sankey diagram, adding migration context without a separate chart.

The structure of the third version was kept in the final one (Appendix 1), and the last changes focused on readability. The title “Leaving Portugal pays off, but not equally everywhere” focused on the general gain of emigrating rather than on housing, the core of the work, and described the comparison rather than its result. It was therefore replaced by

one that puts the rent burden first and states the main finding, with a subtitle defining the scenario. The second section title was extended to the destinations of emigration, so that each chart is introduced by the research questions it answers. The most visible change was in colour. The bluish green used until then was not clearly read as a favourable category and, as it covered most of the cards and bands, it made the whole figure darker and less appealing. It was replaced by a lighter and brighter green, more commonly associated with a positive outcome, which also strengthens the contrast with the red. Finally, the rent share was removed from the end labels of the Sankey diagram, as it was already shown in the map, the legend labels of the two charts were unified, the text was enlarged and the authorship was moved to the footer.

Intermediate versions were also shown to family members and friends without a background in data analysis. The idea of counting rent in workdays was easily understood and often seen as the most striking element, while the dot plot and the stacked bars were found harder to read, which reinforced the decision to replace both with the Sankey diagram. Some readers also confused the colours of the wage map (Figures 1 and 2) with those of the threshold and found the pink and red of the waffle charts too similar, which led to the new colour scheme. The final version was also tested with the colour vision deficiency emulator of the colorspace (Zeileis et al., 2020) package for R (R Core Team, 2026), and the green and red categories remained distinguishable in the three simulations and in the desaturated version (Appendices 3a to 3d).

6. Insights and Discussion

The visualization reveals large differences in housing affordability and disposable income across European cities, and shows that higher living costs do not necessarily mean a greater financial burden once wages are taken into account. Regarding RQ1, Lisbon stands out as the least affordable city. A worker earning the average Portuguese wage needs about 11.7 workdays per month to pay their share of the rent, equivalent to 54% of the net wage. Paris (47%) and London (44%) are the only other cities above the 40% threshold, and Madrid comes close with 38%. The most revealing result, however, is that Lisbon's rent is not particularly high by European standards. Bern requires only 3.2 workdays (15%) despite a higher rent, and in Berlin, with the same rent as Lisbon (€1,750 per month), only 6.8. The problem in Lisbon is therefore less the level of rents than the gap between rents and wages. On the other hand, within the same country rents make the difference, as Munich requires 7.9 workdays against Berlin's 6.8 with the same national wage.

Regarding RQ2, the Sankey diagram shows that the gap between Lisbon and the other cities widens at each stage of the month. The average gross wage in Switzerland is about four times the Portuguese one, but after taxes, rent and everyday expenses, a worker in Bern is left with €5,291 against €241 in Lisbon, more than twenty times as much. Taxes vary considerably between countries, from 18% of the gross wage in Switzerland to around 39%

in Belgium and Germany. Their effect is clearest in Brussels and The Hague. Both cities start with almost the same gross wage, but taxes take 39.6% in Belgium against 27.9% in the Netherlands, so Brussels ends the month behind The Hague even though its rent is lower. Everyday expenses also weigh heavily in Lisbon, where average spending on food, transport and communication (€499 per month) is similar to Belgium's and higher than Spain's. Lisbon is therefore the only city where almost nothing is left at the end of the month, while Geneva (€4,691) and Luxembourg (€2,795) leave a wide margin, and even Madrid and Paris, despite their high rents, leave around €900 to €1,000.

Regarding RQ3, the link between migration flows and financial outcome is only partial. Switzerland, the country that receives the most Portuguese emigrants (11,420 in 2025), is also the one where the most money is left at the end of the month. Spain and France, however, receive the second and third largest flows (11,332 and 8,088 in 2024), even though Madrid and Paris leave little more than €900 per month. Luxembourg, on the other hand, offers the second-best outcome but receives only 3,147 Portuguese per year. This suggests that the choice of destination cannot be explained by financial considerations alone, and that factors such as geographical proximity, language and existing Portuguese communities also play a role.

Taken together, the results support the title of the visualization. Lisbon combines the heaviest rent burden with the least money left at the end of the month, so moving to any of the destinations considered improves the financial situation of the average worker, although the gain ranges from a few hundred euros per month in Madrid or Paris to several thousand in Switzerland.

7. Limitations

The results should be read as a standardised comparison between cities rather than as an exact description of the living conditions of any individual worker. Most limitations result from the use of harmonised OECD and Eurostat datasets, which make countries comparable at the cost of local detail.

The most important limitation concerns wages. Wages and everyday expenses are national averages, while rents are specific to each city. In capitals such as Lisbon, Paris or London, wages tend to be higher than the national average, so the burden of rent may be overestimated for these cities. On the other hand, the average wage is pulled upwards by high earners and is usually above what a typical worker, or a recently arrived emigrant, actually earns, so for them the figure underestimates the workdays needed to pay rent and overestimates what is left.

Rent data raise a similar question, since they come from a Eurostat series used to adjust the salaries of EU staff, which refers to dwellings of a relatively high standard (Eurostat, n.d.) and may inflate rents where the gap between high-end and average housing

is wide, as in Lisbon. Sharing the flat between two people keeps the comparison consistent, but it does not represent single-person households or families. Everyday expenses are also an approximation, as they are averages per inhabitant, including children and people without income, and cover only food, transport and communication. Electricity and gas could not be included because Eurostat does not publish them, and health, education and leisure were not considered, so the amount left at the end of the month is an upper estimate. In the same way, the 40% threshold is applied to rent alone, while Eurostat uses total housing costs and household income (Eurostat, 2026d), so it works as a reference rather than as the official measure.

The data also combine different reference years. Wages, taxes, rents and most emigration figures refer to 2025, while everyday expenses and emigration to Spain and France refer to 2024, and each country records foreign arrivals in its own way. For RQ3, the figure therefore shows whether migration flows and financial outcomes coincide, not why people choose a destination, which would require data such as surveys of Portuguese emigrants.

The design choices also involve trade-offs. Expressing rent in workdays makes cities comparable, but it hides the rent itself, so the map does not show that Lisbon's rent equals Berlin's. Placing the cards on a map adds geographical context, but it makes the ranking harder to read than in a sorted bar chart. The two colour categories make the 40% threshold visible at a glance, but colour alone places Madrid (38%) in the same group as Bern (15%), although Madrid is close to the limit. In the Sankey diagram, the crossing bands show changes in position, but distant bands are hard to compare and Lisbon's band becomes very thin in the last stage.

Team Members' Contributions

The project theme, the choice of datasets and the main design decisions were discussed by all three team members. Leonor Carvalho prepared and processed the data and developed the Sankey diagram, including the additional data it required. Mariana Pinho developed the map with waffle charts and refined its visual representation. Seyed Mirheydari was responsible for the visual design, formatting, and presentation of the report, which was written and structured by Leonor and Mariana.

Use of Generative AI

Generative AI tools, particularly ChatGPT and Claude, were used to support R programming and code debugging, as well as to improve the structure and clarity of the report. Dataset selection, methodological assumptions, design decisions and the interpretation of the results remained the responsibility of the team, and all suggestions were reviewed and adapted to the project's objectives.

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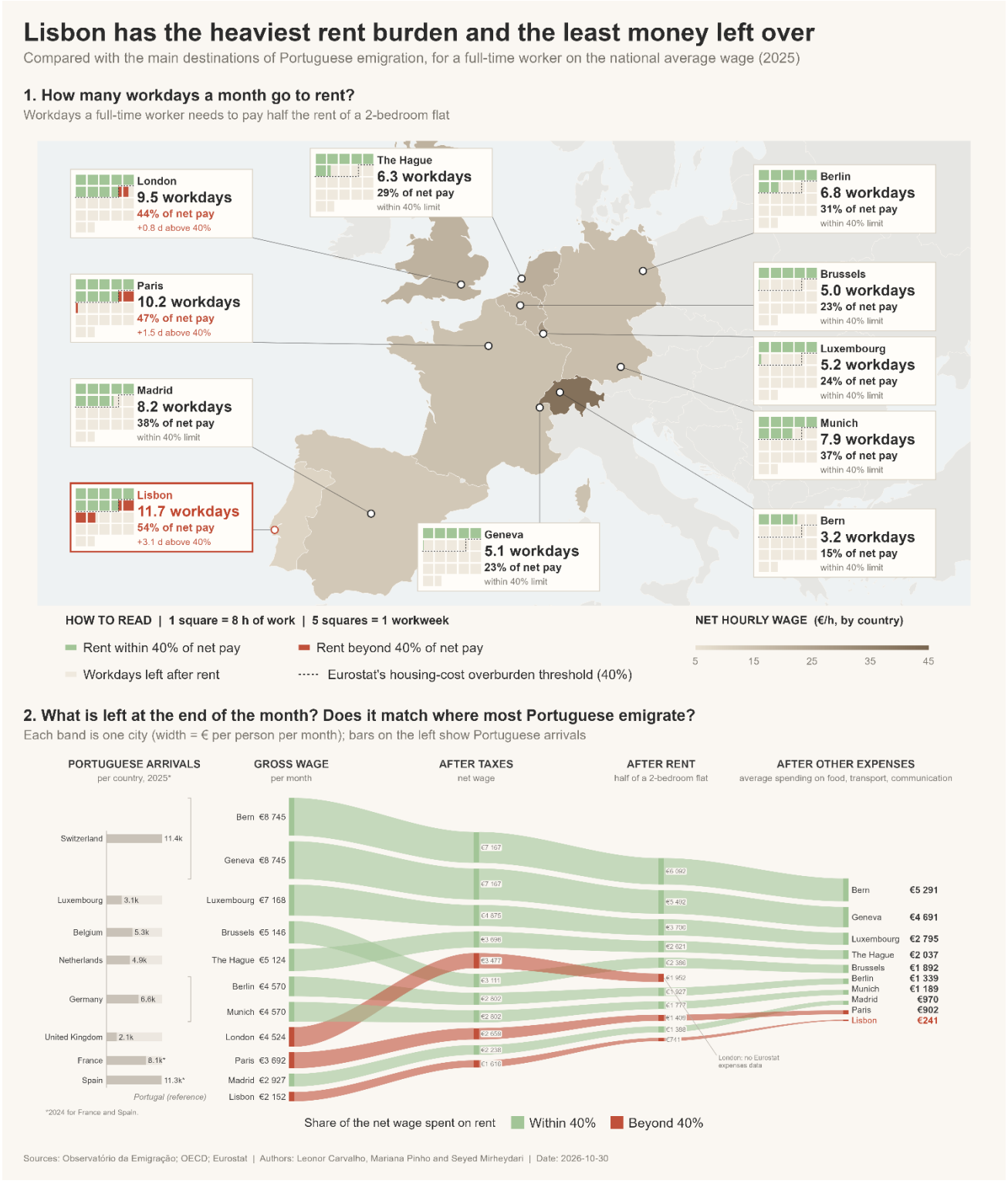
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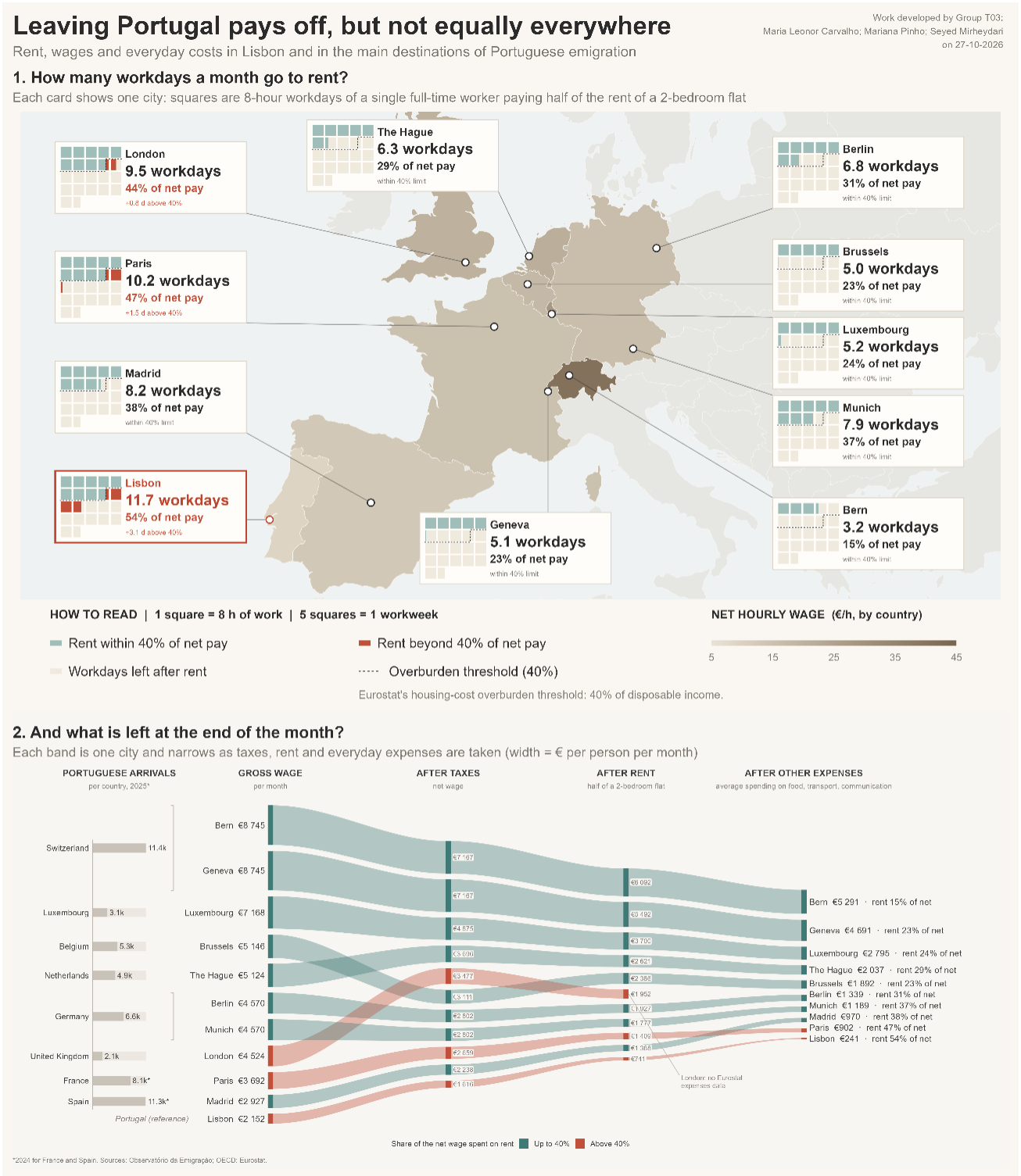
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Appendix

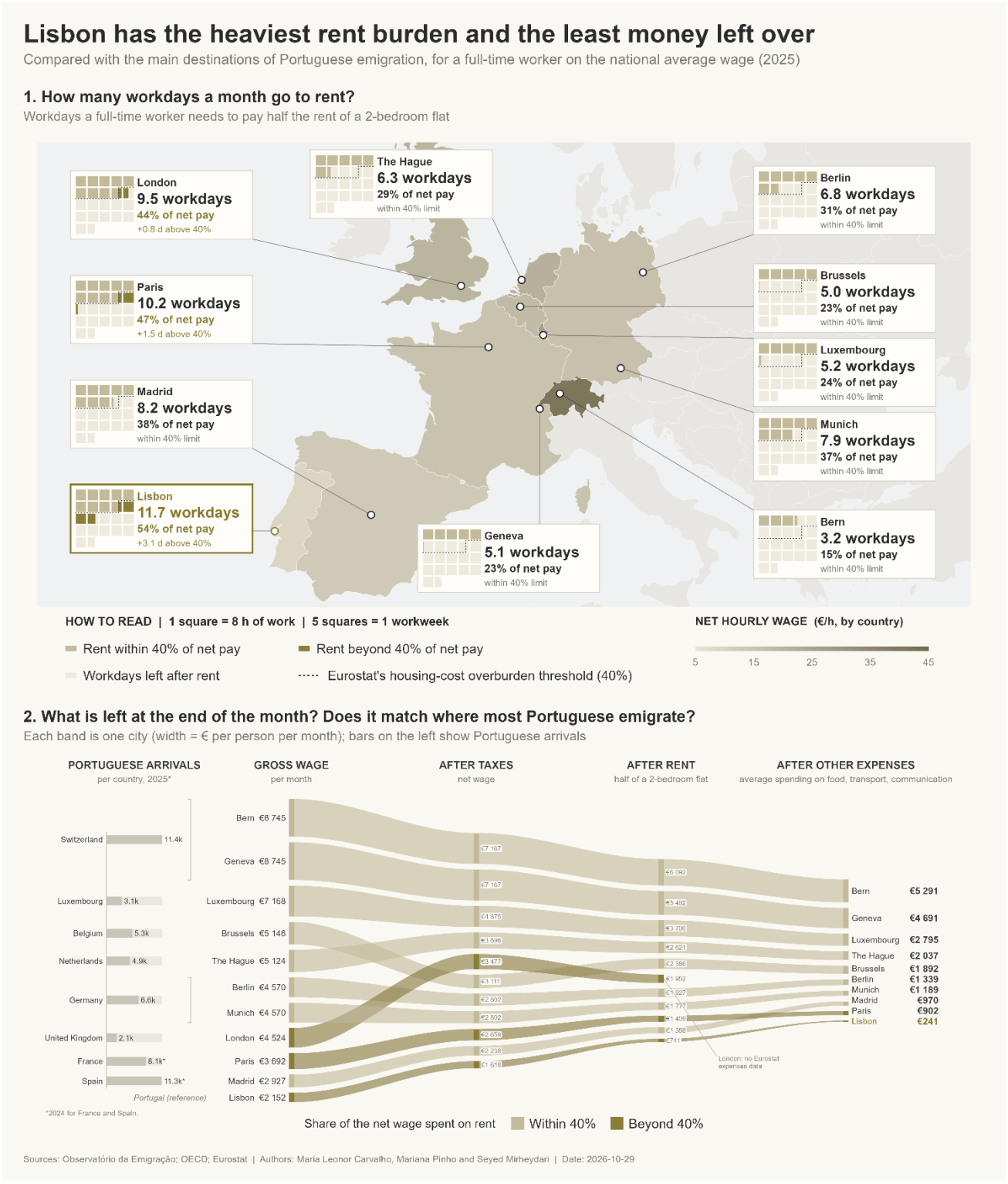
The visualizations are reproduced here at full page size, as their level of detail would make them difficult to read within the main text. The high-resolution file, exported at 300 dpi with ragg (Pedersen & Shemanarev, 2026) (T03_visualization.png) is also submitted separately.



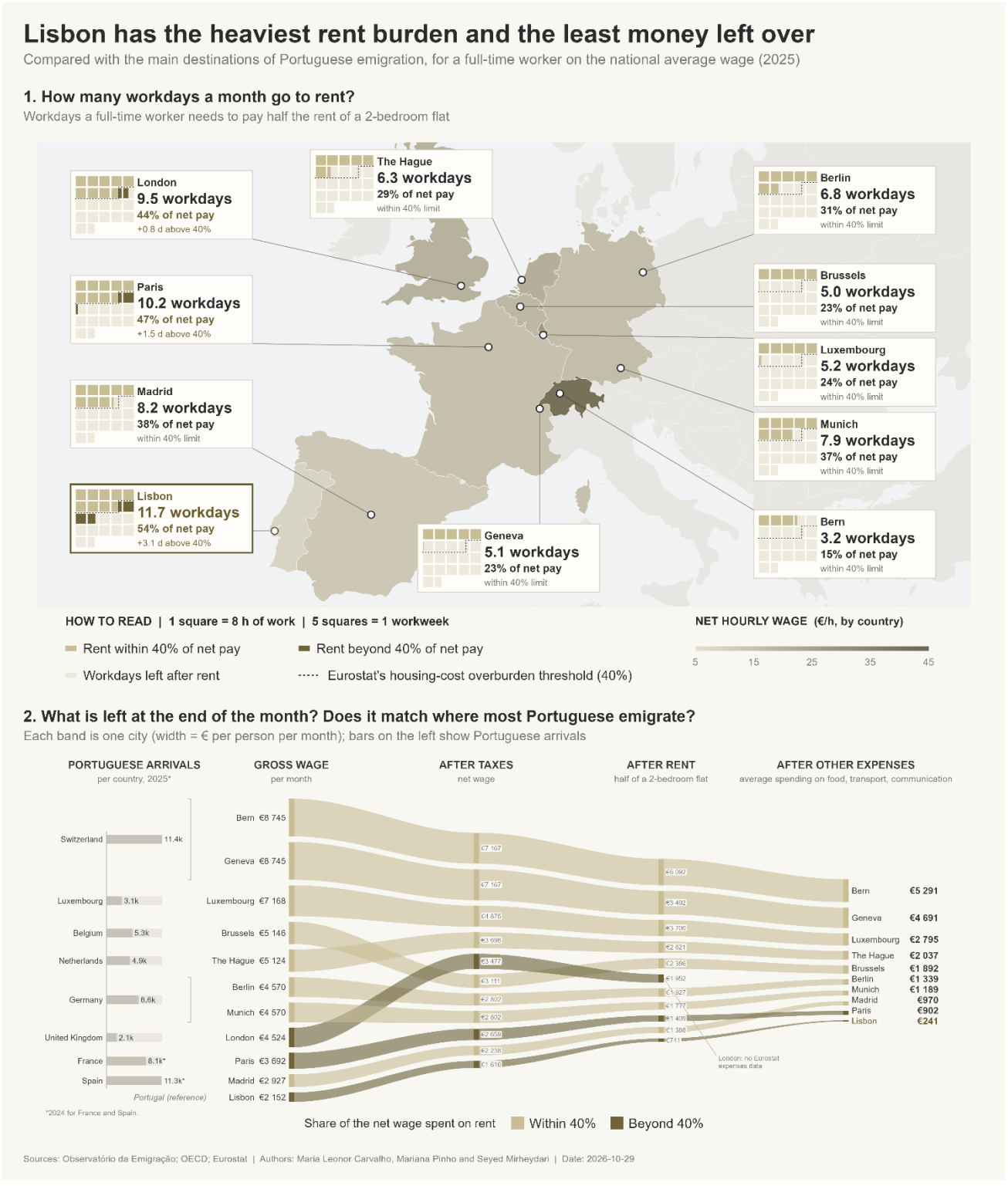
Appendix 1: Final visualization



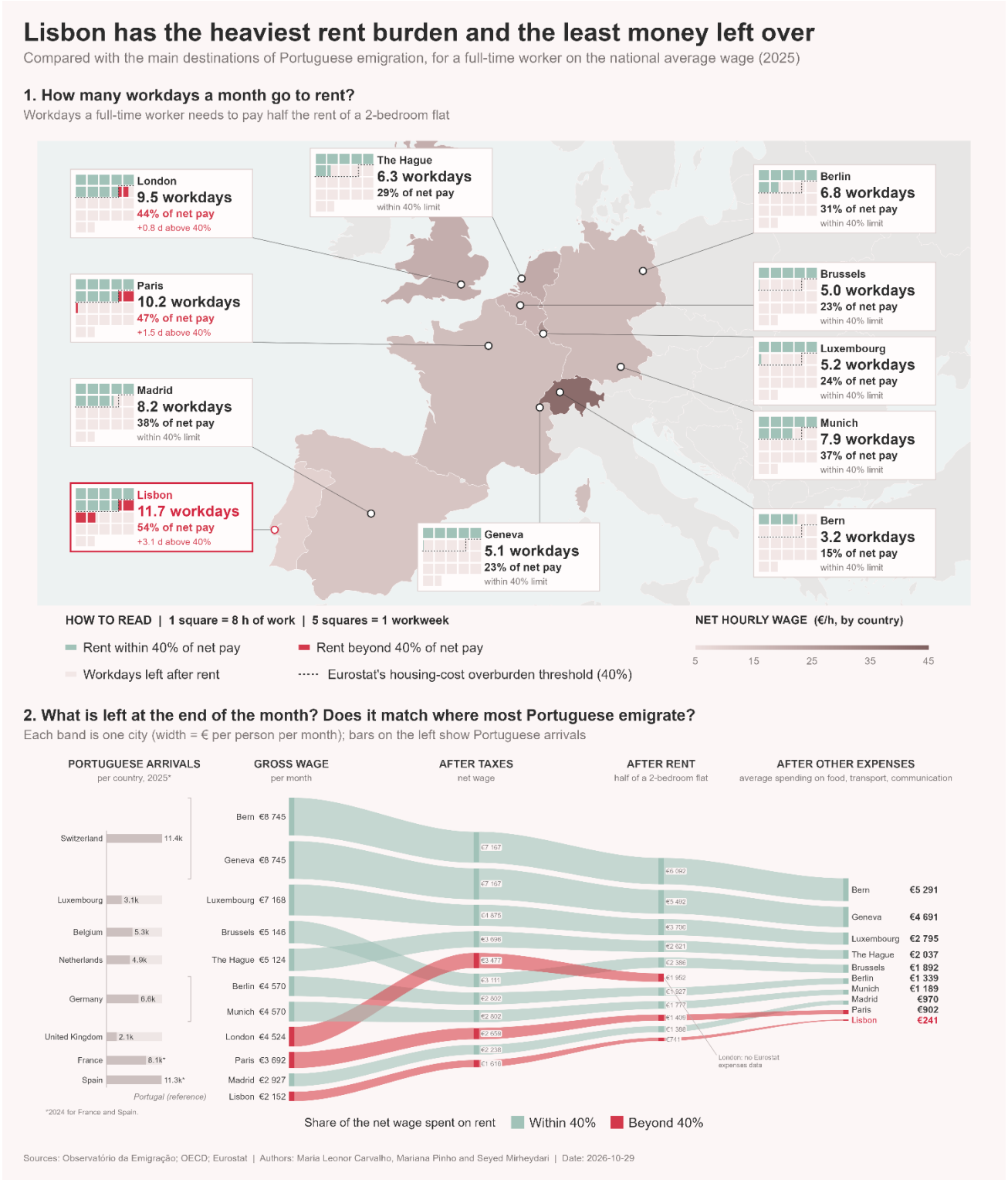
Appendix 2: Third intermediate version



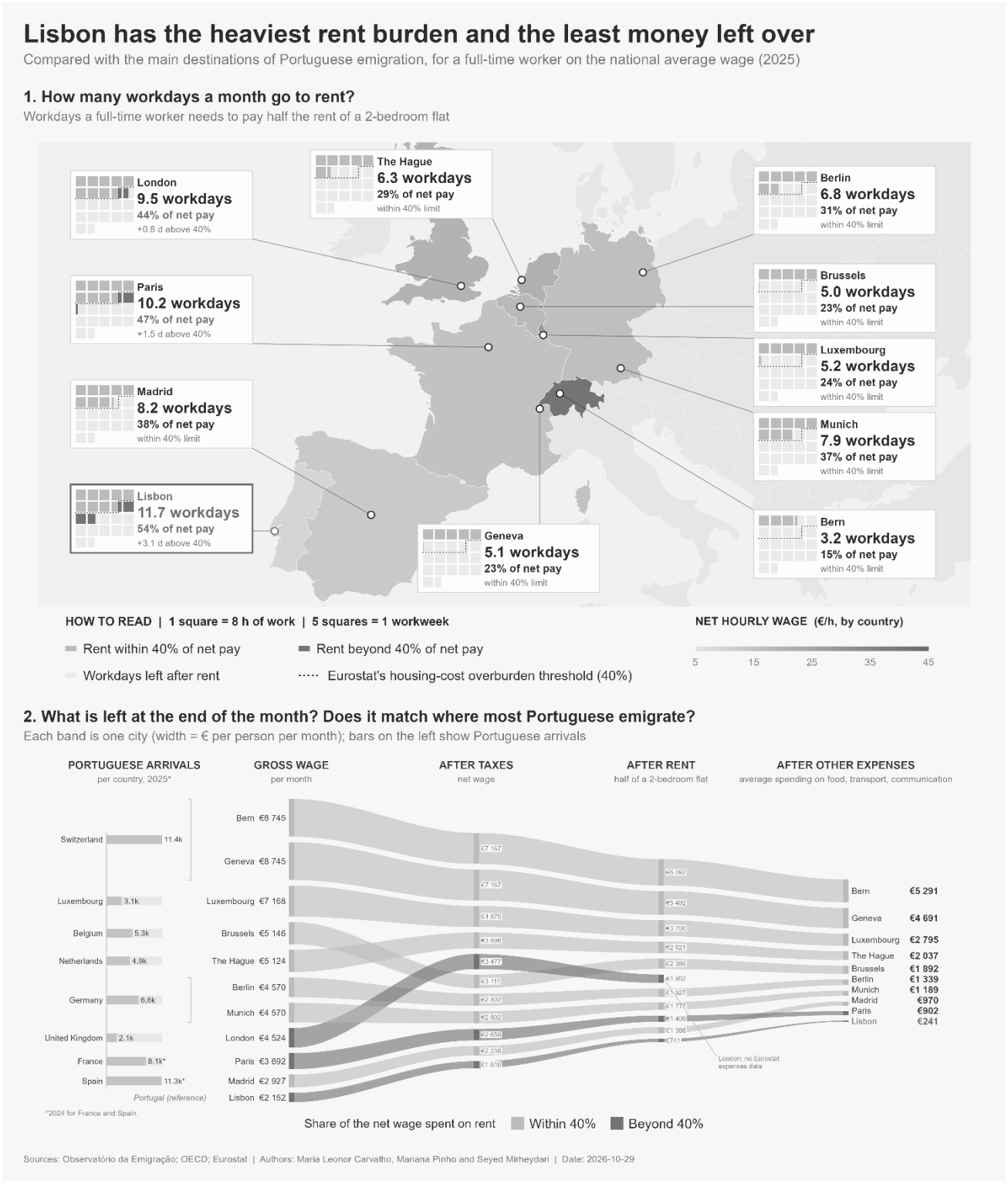
Appendix 3a: Final visualization simulated for deuteranopia



Appendix 3b: Final visualization simulated for protanopia



Appendix 3c: Final visualization simulated for tritanopia



Appendix 3d: Final visualization desaturated (greyscale)